

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

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ESTREL TRIAL Organization

Steering Committee

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Data and Safety Monitoring Board

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Study Medication Production

The study medication was manufactured by Bichsel Apotheke (Laboratorium Dr. G. Bichsel AG, Weissenaustrasse 73, 3800 Unterseen, +41 33 827 80 00, info@bichsel.ch), ensuring compliance with Good Manufacturing Practice (GMP) standards.

Affiliations

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Study Centers

Acute Study Centers (n=13)

University Hospital Basel

University Hospital Zurich

Cantonal Hospital Fribourg (HFR)

University Hospital Bern (Inselspital)

University Hospital Lausanne (CHUV)

Cantonal Hospital St. Gallen

Cantonal Hospital Münsterlingen

Cantonal Hospital Winterthur

Cantonal Hospital Baden

Cantonal Hospital Graubünden

Cantonal Hospital Aarau

Hirslanden Clinic, Zurich

Cantonal Hospital Sion

Rehabilitation Study Centers (n=11)

University Geriatric Medicine Felix Platter, Basel

Rheinfelden Rehabilitation Center

HFR Meyriez-Murten Rehabilitation Clinic

Zihlschlacht Rehabilitation Clinic

Zurich Rehabilitation Center, Lengg

Cereneo — Center for Neurology & Rehabilitation (Vitznau)

Bad Zurzach Rehabilitation Clinic

Valens Rehabilitation Clinic

Zurich Rehabilitation Center, Wald

Rheinburg Rehabilitation Clinic Walzenhausen

Sion Rehabilitation Clinic, Sion

Additional Details About Statistical Analyses

Statistical Software Used and Quality Control

All statistical analyses were conducted using R (version 4.3.1, released on June 16, 2023). The following R packages were utilized for data processing, statistical modeling, and visualization:

Core Packages:

- Data Manipulation & Wrangling: dplyr (1.1.4), data.table (1.14.8), tidyr (1.3.0), forcats (1.0.0), plyr (1.8.9), purrr (1.0.2), tibble (3.2.1), tidyselect (1.2.0)
- Statistical Modeling & Inference: lme4 (1.1-35.1), MASS (7.3-60), Matrix (1.6-3), mice (3.16.0), mitml (0.4-5), glmnet (4.1-8), survey (4.2-1), survival (3.5-7)
- Machine Learning & Classification: e1071 (1.7-13), nnet (7.3-19), rpart (4.1.19), class (7.3-22)
- Graphics & Visualization: ggplot2 (3.4.4), scales (1.2.1), grid (4.3.1), gtable (0.3.4), colorspace (2.1-0), munsell (0.5.0)
- Quality Control & Utilities: assertthat (0.2.1), boot (1.3-28), DBI (1.1.3), digest (0.6.36), stringr (1.5.1), stringi (1.8.4), tools (4.3.1)

Quality control measures involved independently verifying the statistician's code, cross-checking key dataset elements for accuracy, and validating the consistency of tables and figures with the underlying data.

Multiple Imputation Details

We used 100 multiple imputations using chained equations. The imputed values, that have been generated by the imputation model, were displayed in plots. Thereafter the model was fit on each imputed data set. Estimates and variances were combined using Rubins rules to give a single estimate and variance for each parameter of interest across all imputations. For the Fugl-Meyer Motor Assessment the missing or incomplete data was imputed on the item level, whereas for the secondary outcomes the summary scores were imputed. Outcomes for potential harm were not imputed.

Deviations From the Statistical Analysis Plan

We had two deviations from the Statistical Analysis Plan:

- 1.) The modified Rankin Scale was not imputed for participants who passed away before the three-month assessment but was set at 6 as defined by the scale.
- 2.) Missing data for the subjective Patient-reported Assessment of Relevance of Motor Improvement (PRAI) were not imputed.
- 3.) A spelling error was identified in the Statistical Analysis Plan regarding the estimated adjusted mean difference for the secondary outcome 'affected side'. Initially stated as the affected side, the calculation was performed for the unaffected side, which is the correct approach.

Patient-reported assessment of the relevance of motor improvement

Details about the patient-reported assessment of the relevance of motor improvement:

1) Haben Sie seit dem letzten Studienbesuch eine Verbesserung der Defizite verspürt, die durch den Schlaganfall verursacht wurden?

Have you noticed an improvement in the deficits caused by the stroke since your last study visit?

2) Wenn ja, ist diese Verbesserung für Sie im Alltag bedeutsam?

If yes, is this improvement meaningful for you in daily life?

Definition of the Additional Distal Motor Function Item

The patient's hand is held up at the forearm by the examiner and patient is asked to extend his/her fingers as much as possible. If the patient can't or doesn't extend the fingers the examiner places the fingers in full extension and observes for any flexion movement for 5 seconds. The patient's first attempts only are graded. Repetition of the instructions or of the testing is prohibited.

0 = Normal (No flexion after 5 seconds)

1 = At least some extension after 5 seconds, but not fully extended. Any movement of the fingers which is not command is not scored.

2 = No voluntary extension after 5 seconds. Movements of the fingers at another time are not scored.

According to REF:

Egelko A, Kirchoff-Torres KF, Ramaswamy S, et al. Including Distal Motor Function within the NIHSS: Correlation with Motor Arm Function and IV rt-PA Treatment Response. *J Stroke Cerebrovasc Dis* 2021; 30(11): 106046.

Leira EC, Coffey CS, Jorge RE, et al. The NIHSS supplementary motor scale: a valid tool for multidisciplinary recovery trials. *Cerebrovasc Dis* 2013; 36(1): 69-73.

Supplemental Tables and Figures

eTable 1. Detailed Baseline Demographic and Clinical Characteristics Stratified by Levodopa and Placebo Groups

	Overall	Placebo	Levodopa
N	610	303	307
Median age [IQR] - yr	73 [64-82]	74 [64-84]	72 [62-81]
Female Sex– No. (%)	252 (41.3)	133 (43.9)	119 (38.8)
Median Body Mass Index [IQR]	25.7 [23.4 28.7]	25.3 [23.1, 28.1]	26.0 [23.5, 29.1]
Medical history – No. (%)			
Hypertension	435 (71.3)	213 (70.3)	222 (72.3)
Diabetes	144 (23.6)	64 (21.1)	80 (26.1)
Hyperlipidemia	291 (47.7)	152 (50.2)	139 (45.3)
Smoking	127 (20.8)	59 (19.5)	68 (22.1)
Atrial fibrillation	95 (15.6)	61 (20.1)	34 (11.1)
Coronary heart disease	99 (16.2)	47 (15.5)	52 (16.9)
Dementia	20 (3.3)	9 (3.0)	11 (3.6)
Movement disorder	11 (1.8)	5 (1.7)	6 (2.0)
Depression	41 (6.7)	18 (5.9)	23 (7.5)
Prior stroke – No. (%)			
Ischemic stroke	82 (13.4)	43 (14.2)	39 (12.7)
Hemorrhagic stroke	11 (1.8)	6 (2.0)	5 (1.6)
Prior median mRS [IQR]	0.0 [0.0, 1.0]	0.0 [0.0, 1.0]	0.0 [0.0, 0.0]
Prestroke living situation – No. (%)			
Home	596 (97.7)	294 (97.0)	302 (98.4)
Nursing home	9 (1.5)	5 (1.7)	4 (1.3)
Other	5 (0.8)	4 (1.3)	1 (0.3)
Stroke ischemic – No. (%)	519 (85.1)	259 (85.5)	263 (85.7)
Stroke hemorrhagic – No. (%)	91 (14.9)	44 (14.5)	44 (14.3)
MCA affected – No. (%)	460 (75.4)	230 (75.9)	230 (74.9)
Bilateral	16 (2.6)	6 (2.0)	10 (3.3)
Right	278 (45.6)	136 (44.9)	142 (46.3)
Left	166 (27.2)	88 (29.0)	78 (25.4)
ACA affected – No. (%)	68 (11.1)	31 (10.2)	37 (12.1)
Bilateral	4 (0.7)	2 (0.7)	2 (0.7)
Right	23 (3.8)	11 (3.6)	12 (3.9)
Left	41 (6.7)	18 (5.9)	23 (7.5)
PCA affected – No. (%)	60 (9.8)	33 (10.9)	27 (8.8)
Bilateral	8 (1.3)	5 (1.7)	3 (1.0)

Right	35 (5.7)	20 (6.6)	15 (4.9)
Left	17 (2.8)	8 (2.6)	9 (2.9)
Vertebrobasilar affected – No. (%)	108 (17.7)	45 (14.9)	63 (20.5)
Bilateral	4 (0.7)	1 (0.3)	3 (1.0)
Right	59 (9.7)	28 (9.2)	31 (10.1)
Left	45 (7.4)	16 (5.3)	29 (9.4)
Median FMA Total, affected side, at randomization [IQR]	34.0 [14.0, 54.0]	33.0 [13.0, 52.8]	35.0 [14.8, 55.3]
Median FMA Upper, affected side, at randomization [IQR]	18.0 [6.0, 34.0]	17.5 [6.0, 34.0]	20.0 [6.0, 35.0]
Median FMA Lower, affected side, at randomization [IQR]	14.0 [6.5, 21.0]	14.0 [7.0, 21.0]	15.0 [6.0, 21.0]
Median NIHSS at randomization [IQR]	8.2 (3.9)	8.3 (3.8)	8.2 (3.9)
Median mRS at randomization [IQR]	4 [4, 5]	4 [4, 5]	4 [4, 5]
Median time from stroke onset to randomization [IQR]	7 [5, 10]	8 [5, 10]	7 [5-11]
Acute Stroke Treatment – No. (%)			
Intravenous thrombolysis	120 (19.7)	66 (21.8)	54 (17.6)
Bridging therapy	33 (5.4)	13 (4.3)	20 (6.5)
Endovascular therapy	81 (13.3)	44 (14.5)	37 (12.1)

Legend: This table summarizes the baseline characteristics of the study participants. IQR = Interquartile Range, MCA = Middle Cerebral Artery, ACA =Anterior Cerebral Artery, PCA = Posterior Cerebral Artery, mRS = modified Rankin Scale, NIHSS = National Health Institutes Stroke Scale, FMA = Fugl-Meyer Motor Assessment.

eTable 2. Estimand Definitions and Results

Population Definitions for the Estimands

Populations	Definition
Full analysis set	All randomized participants were included. Multiple imputation using chained equations was applied to impute missing FMA values among participants who survived up to the three-month assessment. Participants who died before the three-month assessment were excluded from the analysis.
Modified intention to treat set	All participants in the full analysis set with a complete FMA at the three-month assessment.
Treatment	Levodopa in combination with standardized rehabilitation vs placebo in combination with standardized rehabilitation.

Results

Similar differences in FMA total scores at 3 months between both treatment groups were present for the eight estimands: (estimand 2) including participants with a complete 3-month FMA assessment (-0.39, 95% CI, -3.33 to 2.56), (estimand 3) including and imputing participants who died before the 3-month assessment (-0.98, 95% CI, -3.77 to 1.81), (estimand 4) applying composite strategy using win-ratios resulting in an odds ratio of 1.06 (95% CI, 0.86 to 1.25), (estimand 5) excluding participants with serious adverse events causing motor impairment (-1.41, 95% CI, -4.31 to 1.49), (estimand 6) excluding participants with intake of less than 80% of the study medication (-0.57, 95% CI, -3.73 to 2.58), (estimand 7) excluding participants with fewer rehabilitation therapies than foreseen (0.19, 95% CI, -3.11 to 3.49), and (estimand 8) excluding participants with 3-month assessments outside the predefined time range (-0.33, 95% CI -3.31 to 2.65).

Name	Population	Endpoint	Population-level summary	Intercurrent Events	Estimated Effect of Levodopa: Mean Difference on FMA, [CI]	Number of participants
Estimand 1: Primary Endpoint , intention-to-treat principle						
Treatment policy strategy	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	According to the intention-to-treat principle, all participants were analyzed according to randomization, irrespective of the presence or absence of intercurrent events.	-0.90 [-3.78 – 1.98]	582
Estimand 2						
Treatment policy strategy II	Modified intention to treat set	FMA at the three-month visit	Mean difference between the two treatment arms	According to the intention-to-treat principle, all participants were analyzed according to randomization irrespective of the presence or absence of intercurrent events.	-0.39 [-3.33 - 2.56]	551
Estimand 3						
Treatment policy strategy III	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	FMA of participants who died were imputed.	-0.98 [-3.77 - 1.81]	610
Estimand 4						
Composite strategy using win-ratios	Full analysis set	Death and FMA measured at the three-month visit	Difference in win-ratio at the three-month visit, win ratio [95% CI]	According to the intention-to-treat principle, all participants were analyzed according to randomization irrespective of the presence or absence of intercurrent events.	1.06 [0.86 - 1.26]	610
Estimand 5						
Hypothetical strategy	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	Excluding participants who had a serious adverse event leading to a motor impairing event prior to the three-month visit (e.g., recurrent strokes, fractures, comparable complications)	-1.41 [-4.31 - 1.49]	545
Estimand 6						
On treatment strategy	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	Excluding participants who took less than 80% of the study medication	-0.57 [-3.73 - 2.58]	496
Estimand 7						
On treatment strategy II	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	Excluding participants who had at least one week with fewer than five rehabilitation sessions per week during a period of 4 weeks	0.19 [-3.11 – 3.49]	450
Estimand 8						
On treatment strategy III	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	Excluding participants for whom the three-month visit was out of the window	-0.33 [-3.31 – 2.65]	524

eTable 3. Post Hoc Estimand Analyses

Name	Population	Endpoint	Population-level summary	Intercurrent Events	Estimated Effect of Levodopa: Mean Difference on FMA, [CI]	Number of participants
Estimand 9: Primary Endpoint calculated with the change in FMA from baseline to the three-month visit , intention-to-treat principle						
Treatment policy strategy	Full analysis set	Change in FMA from baseline to the three-months visit	Mean difference between the two treatment arms	According to the intention-to-treat principle, all participants were analyzed according to randomization, irrespective of the presence or absence of intercurrent events.	-0.90 [-3.78 – 1.98]	582
Estimand 10						
Treatment policy strategy IV	All randomized participants	FMA at the three-month visit	Mean difference between the two treatment arms	According to the intention-to-treat principle, all participants were analyzed according to randomization irrespective of the presence or absence of intercurrent events. Participants who died before the three-month assessment were assigned the worst possible FMA outcome, corresponding to an FMA score of 0.	-0.03 [-3.27 – 3.21]	610
Estimand 11						
On treatment strategy IV	Full analysis set	FMA at the three-month visit	Mean difference between the two treatment arms	Excluding participants who had at least one week with fewer than five rehabilitation sessions per week during a period of 4 weeks and who took less than 80% of the study medication.	0.32 [-3.25 – 3.89]	395

eTable 4. Secondary Outcome Results Stratified by Levodopa and Placebo Groups

	Placebo	Levodopa	SMD	Estimated Effect of Levodopa: Mean Difference on FMA, [CI]	Number of participants
FMA total score, unaffected side, 3 months, mean (SD)	89.14 (10.32)	89.67 (9.87)	0.052	0.63 [-0.72 – 1.99]	582
FMA upper, affected side, 3 months, mean (SD)	39.12 (19.19)	39.20 (19.30)	0.004	-0.73 [-2.97 – 1.50]	582
FMA lower, affected side, 3 months, mean (SD)	22.88 (6.90)	22.91 (7.28)	0.003	-0.13 [-1.02 – 0.77]	582
FMA total score, affected side, 5 weeks, mean (SD)	56.27 (25.20)	57.37 (26.70)	0.042	0.02 [-2.81 - 2.84]	582
NIHSS, 3 months, mean (SD)	4.20 (3.37)	4.02 (3.39)	0.054	-0.14 [-0.61 – 0.33]	582
PROMIS 29, 3 months, mean (SD)	65.11 (13.79)	64.74 (14.33)	0.026	-0.37 [-3.34 – 2.61]	582
PROMIS 10, 3 months, mean (SD)	29.87 (5.74)	30.04 (5.73)	0.031	0.18 [-0.98 – 1.33]	582
mRS, 3 months, median [IQR]	3.00 [2.00, 4.00]	3.00 [2.00, 4.00]	0.034	0.93 [0.69 – 1.23]	596
Rivermead Mobility Index, 3 months, mean (SD)	9.89 (4.73)	9.70 (4.85)	0.041	-0.33 [-1.04 – 0.37]	582
PRAI; no (relevant) improvement, 3 months, N. (%)	51 (18.89) (n = 270)	51 (18.48) (n = 276)	-	-	546

Legend: FMA = Fugl-Meyer Motor Assessment, PROMIS 10 and 29 = Patient-Reported Outcomes Measurement Information System, NIHSS = National Institutes Stroke Score, mRS = modified Rankin Scale, PRAI = Patient-reported assessment of relevance of motor improvement, SMD = Standardized Mean Difference, CI = Confidence interval, IQR = Interquartile range.

eTable 5. Details of Serious Adverse Events Stratified by Levodopa and Placebo Groups

	Overall	Placebo	Levodopa
n	255	129	126
Participants died	28	17	11
Hospitalization	212	102	110
Permanent or significant incapacity or disability	23	13	10
Life-threatening	16	10	6
SAE Severity – No. (%)			
severe (daily activities impossible)	103	54	49
moderate (interferes with daily activities)	106	54	52
mild (tolerable)	46	21	25
SAE Outcome – No. (%)			
continuing	6	4	2
recovered/resolved with sequelae	76	40	36
study participant died	29	18	11
unknown	0	0	0
recovered/resolved	142	65	77
other	2	2	0

eTable 6. Details of Prespecified Adverse Events of Interest Stratified by Levodopa and Placebo Groups

	Overall	Placebo	Levodopa
n	145	67	79
AE intensity – No. (%)			
mild	58	29	29
moderate	86	37	49
severe	2	1	1
AE outcome – No. (%)			
ongoing	1	0	1
resolved with sequel	29	13	16
resolved without sequel	116	54	62
Drug relation – No. (%)			
certain	2	1	1
possible	66	32	34
Probably	23	7	16
Unassessable	2	2	0
unlikely	39	15	24
unrelated	14	10	4

Legend: AE = Adverse event. Intensities: mild = tolerable, moderate = interferes with daily activities, severe = daily activities impossible. Drug relation: probably = likely, unassessable = unclassifiable.

eTable 7. Model including interaction between baseline FMA and treatment group, with baseline FMA

term	Estimate	CI	p
(Intercept)	62.51	[60.45;64.57]	<0.001
fma_affected.Visit_1.cent	0.78	[0.69;0.87]	<0.001
medicationLevodopa	-0.90	[-3.78;1.99]	0.542
fma_affected.Visit_1.cent:medicationLevodopa	0.00	[-0.12;0.12]	0.971

Main analysis, Visit 3, including interaction between baseline and follow-up: Estimated treatment effect of Levodopa. When comparing the model with spline (basic spline 3 degrees of freedom) to the model without spline the model with spline was significantly better (F-test $p < 0.01$ for all imputations).

eTable 8. Model without interaction between baseline FMA and treatment group, with baseline FMA modeled using splines (3 degrees of freedom)

term	Estimate	CI	p
(Intercept)	20.09	[14.28;25.90]	<0.001
bs(fma_affected.Visit_1.cent, 3)1	64.73	[46.34;83.13]	<0.001
bs(fma_affected.Visit_1.cent, 3)2	57.52	[43.48;71.56]	<0.001
bs(fma_affected.Visit_1.cent, 3)3	77.49	[61.68;93.31]	<0.001
medicationLevodopa	-0.87	[-3.65;1.92]	0.541

Main analysis, Visit 3, FMA total score, including spline term for FMAA at baseline: Estimated treatment effect of Levodopa. Again there was no evidence for an interaction between FMAA at baseline (with splines) and medication (F-test, p -value > 0.75 for all imputations.)

eTable 9. Prespecified Adverse Events of Interest

Nausea
Vomiting
Taste disturbances
Dry mouth
Anorexia
Arrhythmias
Postural hypotension
Syncope (unconsciousness for a short time as a result of reduced blood flow to the brain)
Drowsiness (including sudden onset of sleep)
Fatigue
Dementia
Psychoses (a distorted perception of reality)
Hallucinations
Confusion
Euphoria
Abnormal dreams
Insomnia
Depression
Anxiety
Dizziness
Dystonia (involuntary contractions)
Dyskinesia (inability to control voluntary movements)
Chorea (sudden twitching of the face and shoulders)

eFigure 1. Eligibility and Exclusion Criteria

Inclusion Criteria

-
1. Acute ischemic or hemorrhagic (i.e. intracerebral hemorrhage excluding subarachnoid hemorrhage and cerebral venous sinus thrombosis) stroke ≤ 7 days prior to randomization
 2. Clinically meaningful hemiparesis (i.e. scoring a total of ≥ 3 points on the following NIH stroke scale score items (i) motor arm, (ii) motor leg, (iii) limb ataxia; a distal arm paresis is equivalent to one of the aforementioned (i-iii))
 3. Time of randomization ≥ 24 -hours since thrombolysis or thrombectomy
 4. In-hospital rehabilitation required
 5. Capable to participate in a standardized rehabilitation therapy
 6. Informed consent of patient or next to kin

Exclusion Criteria

-
1. Age < 18 years
 2. Diagnosis of Parkinson's Disease
 3. Use of Levodopa mandatory according to judgment of treating physician
 4. Inability or unwillingness to comply with study procedures including adherence to study drug intake (orally, or via nasogastric tube or percutaneous endoscopic gastrostomy tube)
 5. Severe aphasia (i.e. unable to follow two-stage-commands)
 6. Previously dependent in the basal activities of daily living (defined as modified Rankin Scale prior to stroke > 3)
 7. Pre-existing hemiparesis
 8. Known hypersensitivity to Levodopa/Carbidopa and other contraindications for Levodopa/Carbidopa as outlined in the summary of product characteristics (as appended to the study protocol).
 9. Woman who are pregnant or breast feeding, or who intend to become pregnant during course of the study. Women of childbearing age must take a pregnancy test to be eligible for the study.
 10. Lack of safe contraception, defined as: Female Participants of childbearing potential, not using and not willing to continue using a medically reliable method of contraception for the entire study duration, such as oral, injectable, or implantable contraceptives, or intrauterine contraceptive devices, or who are not using any other method considered sufficiently reliable by the Investigator in individual cases. Female Participants who are surgically sterilized / hysterectomized or postmenopausal for longer than 2 years are not considered as being of child bearing potential.

eFigure 2. Detailed Study Schedule of the ESTREL Trial

STUDY SCHEDULE	Study phase		Pre-Screening	Enrolment	Interventional phase / main study							Observational phase (clinical)		ESTREL-LONGTERM: extended telephone observational phase				
	Visit		Visit 1		Visit 2				Visit 3	Visit 4	Visit 5	Visit 6	Visit 7	Visit 8	Visit 9	Unscheduled Visit***		
	Main purpose	Eligibility pre-check	Screening, Consent, Enrolment, Randomization	Dose escalation	Transition* to rehab center	Full dose study day	Dose tapering	End of treatment	Primary outcome assessment	Long-term follow up	Long-term follow up	Long-term follow up (telephone visits)	Long-term follow up (telephone visits)	Long-term follow up (telephone visits)	Long-term follow up (telephone visits)	End of study (unscheduled)		
	Timepoint	day -7 to 0	day 0	1-6	day (X)	day 7-34	day 35-39	5 weeks (+/- 7 days)	3 months (+/- 14 days)	6 months (+/- 20 days)	12months (+/- 30 days)	24 months (+/- 30 days)	36 months (+/- 30 days)	48 months (+/- 30 days)	60 months (+/- 30 days)	unscheduled		
	Informed consent	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Randomization	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Dispensing of study drug	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-		
	Group A (Levodopa)	-	X	X	X	X	X	End of treatment	-	-	-	-	-	-	-	-		
	Group B (Placebo)	-	X	X	X	X	X	End of treatment	-	-	-	-	-	-	-	-		
	Distribution of therapy booklet	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-		
Assessments																		
Inclusion-/Exclusion Criteria	X	X	-	-	-	-	-	-	-	-	-	-	-	-	-			
Demographics	X	X	-	-	-	-	-	-	-	-	-	-	-	-	-			
Medical History	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-			
Pregnancy Test (females)	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-			
Physical examination	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Concomitant rehabilitative therapies	-	-	X	X	X	X	X	X	X	X	-	-	-	-	X			
Concomitant medication	-	X	-	X	-	-	X	X	X	X	X	X	X	X	X			
Drug accountability	-	-	-	-	-	-	X	-	-	-	-	-	-	-	[X]			
Blood sampling**	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-			
Outcome measures																		
FMMA (Primary outcome)	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
NIH Stroke Scale Score	X	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Modified Rankin-Scale Score	-	X	-	-	-	-	X	X	X	X	X	X	X	X	X			
PROMIS-10	-	-	-	-	-	-	X	X	X	X	X	X	X	X	X			
PROMIS 29	-	-	-	-	-	-	X	X	X	X	-	-	-	-	X			
Patient reported assessment of improvement	-	-	-	-	-	-	X	X	X	X	X	X	X	X	X			
Rivermead Mobility Index	-	X	-	-	-	-	X	X	X	X	X	X	X	X	X			
Action Research Arm Test**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Box- and Block-Test**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Jamar dynamometer Test**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Motricity Index**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Trunk control**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Functional ambulation categories**	-	X	-	-	-	-	X	X	X	X	X	X	X	X	X			
10-meter walk test**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Montreal Cognitive Assessment**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Daily activity measurement with movement sensors**	-	X	-	-	-	-	X	X	X	X	-	-	-	-	X			
Mortality (all cause)	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X			
Recurrent stroke (any)	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X			
AE/SAE	-	-	X	X	X	X	X	X	-	-	-	-	-	-	[X]			

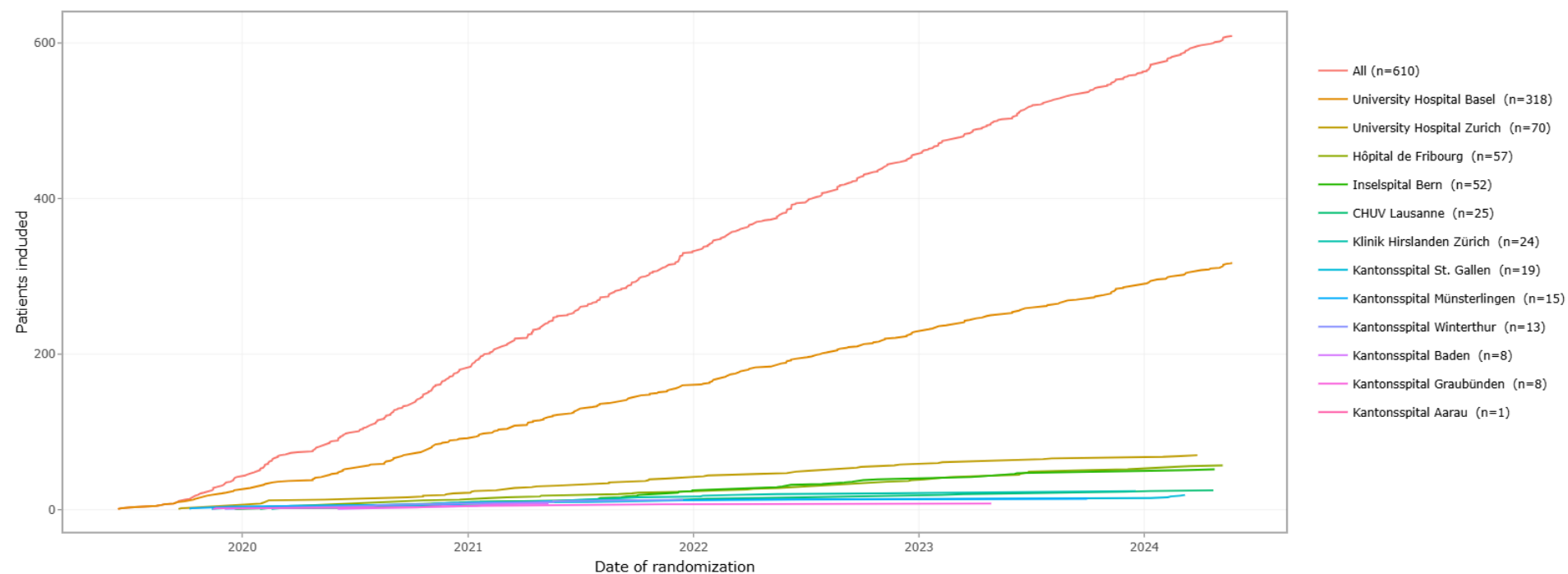
Study Schedule

*Transition: no in-person visit, data collection only, at day of transition of the patient from acute care center to rehabilitation center (any day during treatment phase);

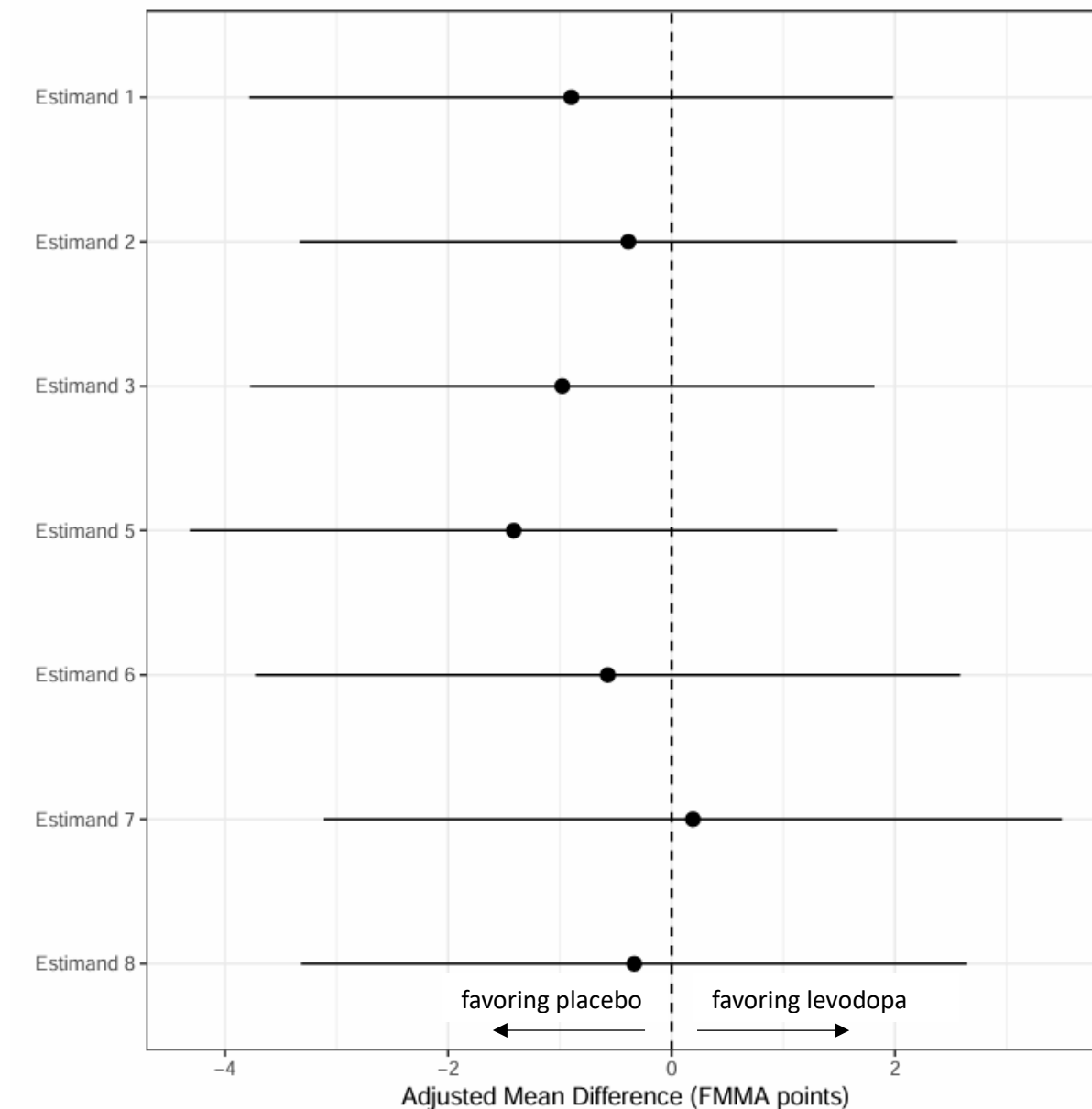
** optional, and in selected centers only, blood sampling may be performed at baseline or any follow up visit.

*** An unscheduled "end of study" visit will be performed in case of discontinuation or withdrawal of consent. [X] Assessment only applicable if the unscheduled visit takes place before end of treatment (regarding drug accountability) or before Visit 3 (regarding AE/SAEs); outcome assessments at this visit will be performed upon participants' approval.

eFigure 3. Cumulative Patient Enrollment by Study Site Over Time



eFigure 4. Forest Plot of FMA Total Score Estimands at Three Months by Treatment Group (Levodopa vs. Placebo)



Legend:

Estimand 1: Excluding participants who died before the 3-month FMA, primary outcome analysis (treatment policy strategy).

Estimand 2: Excluding participants with a missing or incomplete 3-month FMA, complete case analysis (treatment policy strategy II).

Estimand 3: Including and imputing participants who died before the 3-month assessment (treatment policy strategy III).

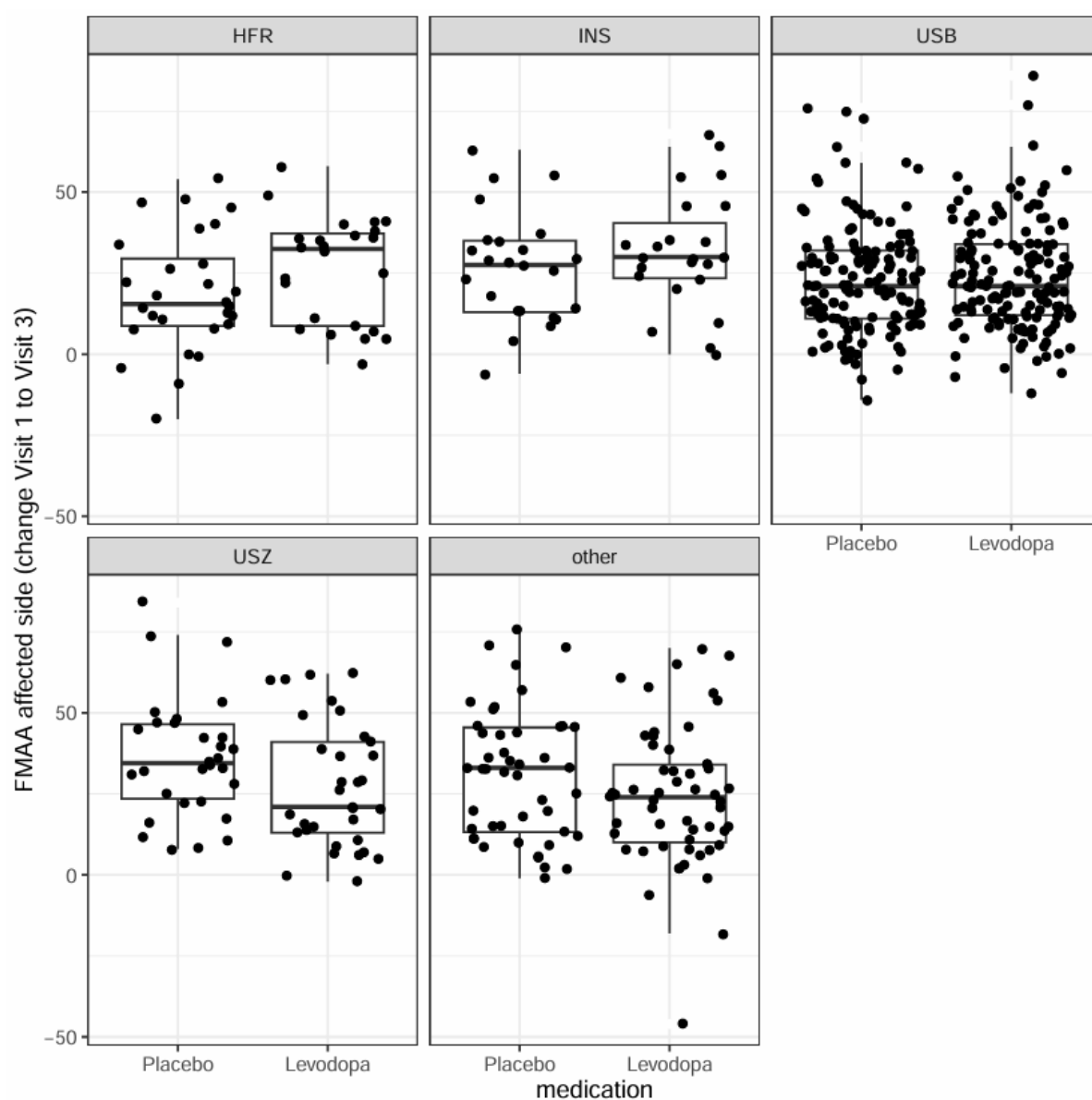
Estimand 5: Excluding participants with serious adverse events causing motor impairment, hypothetical strategy (on treatment strategy).

Estimand 6: Excluding participants with intake of less than 80% of the study medication (on treatment strategy).

Estimand 7: Regarding the amount of rehabilitation training, excluding participants with fewer rehabilitation therapies than foreseen (on treatment strategy II).

Estimand 8: Excluding participants with 3-month assessments outside the predefined time range, measured according to protocol (on treatment strategy III).

eFigure 5. FMA at Three Months by Treatment Arm and Center



Legend: Centers with fewer than 50 recruitments were pooled into the “Other” category.

eFigure 6. Nonlinear association between baseline and three-month FMA (spline model)

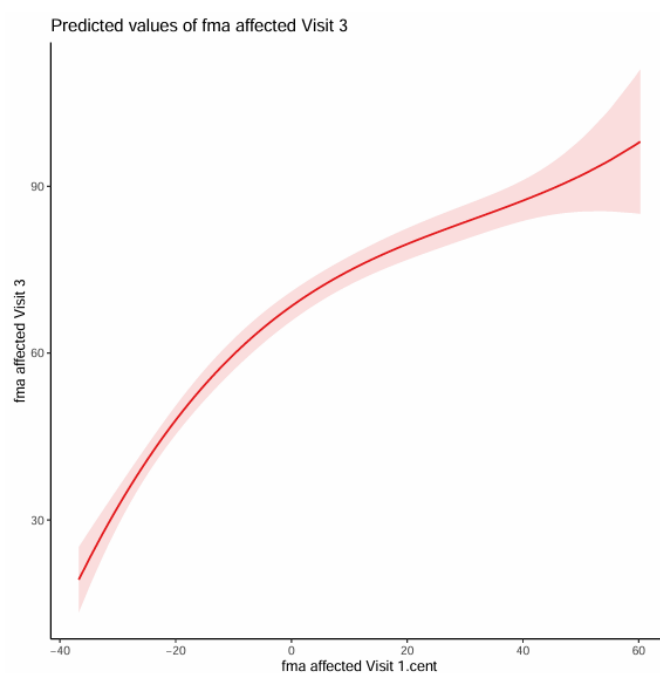
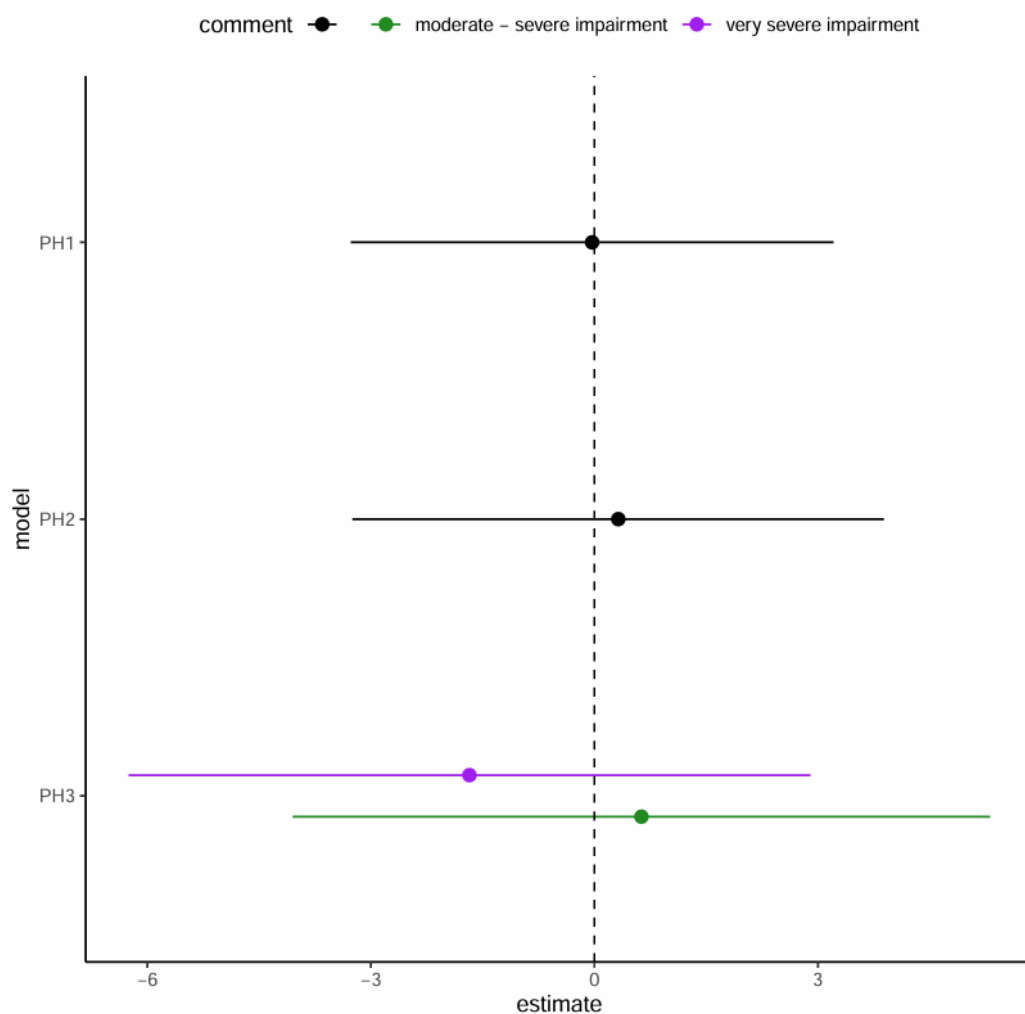


Figure 4: Main estimate including FMAA at baseline as spline: estimated marginal effects.

eFigure 7. Forest Plot of the Post Hoc Estimand and Subgroup Analyses



Legend:

PH1: Post-hoc estimand analysis using the primary analysis model, with missing outcomes for patients who died imputed as 0.

PH2: Post-hoc estimand analysis using the primary analysis model, excluding patients with low baseline intake and minimal documented rehabilitation therapy.

PH3: Subgroup analysis stratifying patients by baseline Fugl-Meyer Motor Assessment (FMA) score: severe impairment (≤ 35 points) vs. mild to moderate impairment (> 35 points).